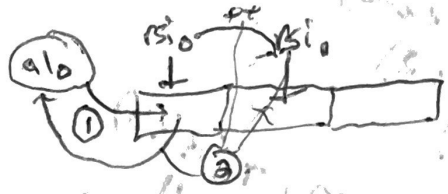
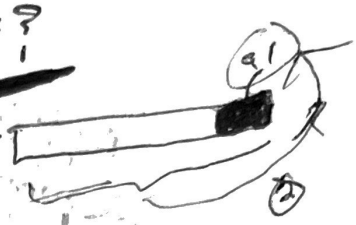
lodsb: [rsi] → al or low-8 bits of rax
& rsi (increment ⟺ DF = 0 default
decrement ⟺ DF = 1)

(default)
DF = 0 ⇒ rsi++
(after each lodsb)



what happens? so al



cumulative xor

is $[rsi+i] = al_0 \oplus \bigoplus_{j=0}^{i} (\text{original byte at})[rsi+j]$

final al is $al_{rax-1} = al_0 \oplus \bigoplus_{j=0}^{rax-1} \text{original byte at } [rsi+j]$

$a_1 \oplus a_0$

```
.loop:
  1 xor        byte [rsi], al
  2 lodsb ]→   iterates over memory (array
    loop       .loop
```

"load string byte"
ex)

loop counter

i = *

[a, b, c]

[a ⊕ *, b, c] i =

a ⊕ * ⊕ b

[a ⊕ *,
a ⊕ * ⊕ *,
a ⊕ b ⊕ c ⊕ *]

al

rcx ⊕ [rsi]
i=0

a ⊕ b

@ function this makes no sense (not obvious) ⇒ data (passed) must hold memory

rcx =

al →